

Vinh Ly

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Summary

Backend and infrastructure engineer with **17 years of experience** building large-scale distributed systems at Google, YouTube, and Whatnot. Comfortable translating deep technical architecture into clear terms for both engineering teams and non-technical stakeholders. Looking to put that communication background to work as a trusted technical voice in sales cycles, helping customers understand how a platform solves their specific problems.

Core Skills

Languages: Python, C++, Java, Go, TypeScript

Distributed Systems: gRPC, REST APIs, Thrift, **microservices**, caching (Redis, Memcached), message queues (Kafka)

Data Stores: Bigtable, Spanner, Postgres, MySQL, Elasticsearch, Firestore

Cloud: GCP (GKE, Compute Engine, BigQuery), AWS (EC2, S3, Lambda)

Observability: Prometheus, Grafana, Datadog, OpenTelemetry

Tooling: Git, Gerrit, GitHub Actions, Terraform, Docker, Kubernetes

Work Experience

Staff Software Engineer | YouTube | Remote | 02/2025 - Present

- Led infrastructure work for the recommendation system serving billions of daily requests. Cut p99 API latency by roughly 35% during peak traffic; helped bring the service to a more stable state after a period of reliability issues.
- Owned the ranking pipeline end-to-end, from feature engineering through serving. Worked closely with ML engineers and product managers to define the offline evaluation framework used to validate ranking changes before launch.

Staff Software Engineer | Whatnot | Hybrid | 05/2024 - 02/2025

- Whatnot is a live commerce marketplace where users buy and sell collectibles through real-time video streams. Joined during a period of rapid user growth and took on backend systems supporting real-time bidding, payment processing, and fraud detection.
- Collaborated with the payments team on the checkout service and reconciliation logic. Helped design the fraud scoring service that reduced chargebacks by a meaningful margin before I moved on.

Staff Software Engineer / Manager | Google | Remote | 03/2015 - 04/2024

- Managed a team of 8 engineers owning search infrastructure and ranking for Google Maps and Shopping. Recruited and hired 4 engineers over 3 years; ran performance reviews and helped shape technical direction across two product areas.
- Improved API reliability for the Places API after a postmortem flagged elevated error rates. Redesigned the batching layer and added stricter circuit-breaking logic; error rate dropped to within SLA.
- Worked on search ranking improvements for Google Shopping, focusing on query understanding and relevance scoring. Spent time on the freshness pipeline that determined how quickly new product listings appeared in search results.

Software Engineer | Google | Remote | 02/2011 - 02/2015

- First stint at Google, working on infrastructure for the Shopping product before it was folded into broader search. Built internal tooling that made it easier for merchant relations teams to manage product catalogue data at scale.

- Developed and maintained backend services handling catalogue ingestion and indexing for millions of product listings. Rewrote a brittle Python import pipeline in Go; throughput roughly doubled and oncall pages dropped significantly.

Software Design Engineer II | Microsoft | Remote | 10/2007 - 12/2011

- Worked on internal developer tools and infrastructure. Built and maintained internal web applications used by several thousand engineers across the company, and contributed to the build and deployment system.
- Led a small effort to migrate a legacy ASP.NET service to a newer framework version. The migration took longer than expected but the outcome was a meaningful reduction in page load times for end users.

Software Design Engineer Intern | Microsoft | Remote | 06/2006 - 08/2006

- Summer internship during college. Worked on a reporting feature for an internal configuration management tool, writing code that shipped to several hundred internal users.

Education

BS Computer Science | UC Irvine | 2007